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Eleftherios (Lef) Ioannidis

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Education

PhD in Computer Science, *University of Pennsylvania*, Philadelphia, PA
Thesis: *"Correct Programs, Executed Correctly: Verifying Specifications and Executions"* (2025).
Advisors: Sebastian Angel & Steve Zdancewic

MEng in Computer Science, *Massachusetts Institute of Technology* (MIT), Cambridge, MA
Thesis: *"Extracting and optimizing low-level bytecode from high-level verified Coq"* (2019).
Advisors: Nickolai Zeldovich & Frans Kaashoek
Adoption: Rocq-to-C++ extraction adopted by Bloomberg (bloomberg/crane) in 2025.

BSc in Computer Science, *Massachusetts Institute of Technology* (MIT), Cambridge, MA
Thesis: *"Parallel annotations for polyhedral optimizations in LLVM"* (2015).
Advisor: Saman Amarasinghe
SuperUROP: Advanced Undergraduate Research in EECS Award (2015).

Employment

2025 - Present — Senior Research Software Engineer, Microsoft Research, Redmond, WA
- Working in the RiSE group, focusing on LLM assisted program verification and proof synthesis.
- AI-assisted **formal verification** of systems using **F$\star$**, **Pulse** and **Verus** proof assistants.
- AI-assisted **bug finding** at scale in the Core **Windows OS**.

Summer 2022 — Research Scientist Intern, Amazon, Automated Reasoning Group, Arlington, VA
- Worked in the **formalization** of the Cedar authorization language and the Cedar validator.
- Implemented a novel **type inference** algorithm for Cedar including singleton and capability types.

2018 - 2019 — Investment Engineer, Bridgewater Associates, Westport, CT
- Created **APIs** used for **big-data** quantitative research, analytics and **visualization**.
- Implemented complex risk-controls and hedging **algorithms** used daily by Trade Generation.
- Taught the **Scala** and **SQL** programming languages to more than 100 traders and engineers.

2016 - 2018 — Founding Engineer & Architect, UnifyID (acquired by Prove), San Francisco, CA
- Designed and implemented a **microservice** based **back-end** on **AWS** (*20 services*).
- Implemented a real-time **Machine Learning** service, for high-throughput inference (*3000 req/sec*).
- Developed certificate management systems and implemented **end-to-end encryption**.
- **Series-A:** $23m raised on a $100m valuation, led by a16z, NEA, StartX and more.
- **1st place**, RSA Innovation Sandbox 2018 Startup Award (Boston, MA).
- **1st place**, SXSW 2018 Startup Competition Award (Austin, TX).
- **Runner-up**, TechCrunch Disrupt 2017 Startup Award (San Francisco, CA).

2015 - 2016 — Software Security Engineer, Apple, Cupertino, CA
- Contributed to the **LLVM compiler**, focus on compiler optimizations for program security.
- Implemented **cryptographic algorithms** for end-to-end encryption and DRM.

Patents & Publications

Aug. 2026. "Proofs Promptly: Proof-Oriented Programming with AI Agents (Experience Report)". In: *Proceedings of the ACM International Conference on Functional Programming (ICFP)*. doi: 10.1145/3828709.

June 2026. "Lazy Validation and Self-Healing for Agentic Programs". In: *Proceedings of the ACM Workshop on Principles of Agentic Engineering (PAgE), Programming Language Design and Implementation (PLDI)*. Boulder, CO, USA. doi: 10.1145/3819802.3820582.

Oct. 2025. "Structural temporal logic for mechanized program verification". In: *Proceedings of the ACM International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA).*
🏆 **Distinguished Artifact Award**.

Mar. 2025. "Authorization Policy Validation". US Patent 12,261,888.

Dec. 2024. "Choice Trees: Representing and reasoning about nondeterministic, recursive, and impure Programs in Coq". In: *Journal of Functional Programming (JFP), Special POPL 2025 Edition.*

Oct. 2024. "Cedar: A New Language for Expressive, Fast, Safe, and Analyzable Authorization". In: *Proceedings of the ACM International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA).*

June 2024. "Reef: Fast Succinct Non-Interactive Zero-Knowledge Regex Proofs". In: *Proceedings of the USENIX Security Symposium (USENIX Security).*

June 2022. "Efficient representation of numerical optimization problems for SNARKs". In: *Proceedings of the USENIX Security Symposium (USENIX Security).*

Mar. 2020. "Privacy-preserving system for machine-learning training data". US Patent 10,601,786.

May 2019. "Extracting and optimizing formally verified code for systems programming". In: *Proceedings of the International Symposium on NASA Formal Methods (NFM).*

Committee Service

2026	PLDI Program Committee
2026	CSF Program Committee
2026	Rocqshop Program Committee